12.3 Matrices

12.3.1 Matrices

'Matrix' is a singular while 'matrices' is a plural form. Matrix is a rectangular array of numbers systematically arranged in rows and columns within brackets. In a matrix, if the number of rows and columns are equal, it is called a square matrix.

12.3.2 Determinants

For every square matrix, there exists a determinant. This determinant is an arrangement of same elements of the corresponding matrix into rows and columns by enclosing vertical lines.

For example,

$\begin{pmatrix} 1 & 3 & 5 \\ 6 & 2 & 4 \\ 7 & 8 & 9 \end{pmatrix}$ is a square matrix of order 3 x 3,then

$\begin{vmatrix} 1 & 3 & 5 \\ 6 & 2 & 4 \\ 7 & 8 & 9 \end{vmatrix}$ is a determinant.

$\begin{pmatrix} 2 & 3 \\ 5 & 7 \end{pmatrix}$ is a square matrix of order 2 x 2, then

$\begin{vmatrix} 2 & 3 \\ 5 & 7 \end{vmatrix}$ is a determinant.

In general, if $A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$ is a matrix then,

$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ is a determinant of the matrix A denoted by $|A|$.

The value of the determinant is expressed as a single number.

Calculation of the value of determinant for a 2 x 2 matrix is shown below

$$\text{If } |A| = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \text{ then } |A| = a_1 b_2 - a_2 b_1$$

Calculation of determinant value for a 3 x 3 matrix is shown below

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1 \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1 \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$$

$$= a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$$

Example: 12.6

Find the value of the determinant for the matrix $A = \begin{pmatrix} 3 & 4 \\ 10 & -2 \end{pmatrix}$

Solution:

Given matrix $A = \begin{pmatrix} 3 & 4 \\ 10 & -2 \end{pmatrix}$ then, the Determinant

$$|A| = \begin{vmatrix} 3 & 4 \\ 10 & -2 \end{vmatrix} = 3(-2) - 10(4)$$

$= -6 - 40 = -46$ is the value of the determinant.

Example: 12.7

Find the value of the determinant of the matrix

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